SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: Computer Networks **COURSE CODE: CSA07**

COURSE OUTCOME COVERED

CO5: *Measure network performance using key metrics with simulation tools*

Assessment Tool Weightage: 20%

QUESTIONS: 40 **Total Marks:40**

Annexure A

Mock Interview

Code of Conduct:

I (Y.Varun kumar reddy/ 192424230) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

QUESTIONS:40 **Total Marks:40**

1. Given a network with high jitter, suggest improvements (BL4)(1)

1. **Prioritize traffic with QoS (Quality of Service)**
   Configure routers to give priority to time-sensitive traffic like voice and video calls over less urgent traffic like file downloads or backups.

2. **Use traffic shaping**
   Smooth out bursts of data so packets move through the network at a more consistent, predictable rate instead of arriving in uneven spikes.

3. **Add a jitter buffer**
   Set up a buffer at the receiving end that temporarily holds incoming packets and releases them at a steady rate, evening out timing differences before playback (commonly used in VoIP and video conferencing systems).

4. **Improve routing**
   Use dedicated or optimized paths (like MPLS or SD-WAN) so packets consistently follow the same, low-variance route instead of jumping between different paths.

5. **Reduce network congestion**
   Upgrade bandwidth or add capacity at bottleneck links so packets aren't queuing up and creating variable delays.

6. **Minimize the number of hops**
   Fewer routers and switches in the path mean fewer points where delay can build up unevenly.

7. **Prefer wired over wireless connections**
   Ethernet connections have much more stable and predictable timing than Wi-Fi, which is prone to interference and signal variation.

8. **Keep hardware and firmware updated**
   Outdated routers, switches, or network drivers can have performance bugs that contribute to inconsistent delays — regular updates help.

9. **Monitor continuously**
   Use network monitoring tools to track jitter, latency, and packet loss in real time so

problems can be caught and fixed early, before they affect users.

Target benchmark: Keep jitter under ~30 ms for smooth VoIP calls, video conferencing, and real-time applications like online gaming.

RUBRICS FOR CALCULATION:

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Conceptual Understanding	30%	Demonstrates thorough understanding; answers all questions accurately	Shows good understanding with few errors	Partial understanding; several incorrect responses	Lacks understanding; majority of answers incorrect
Accuracy of Answers	25%	All answers are correct	Most answers are correct with minor mistakes	Some correct answers; frequent errors	Mostly incorrect answers
Application of Knowledge	20%	Correctly applies concepts to problem-based questions	Applies concepts with minor errors	Limited ability to apply concepts	Unable to apply concepts
Time Management	15%	Completes quiz well within time efficiently	Completes on time with minor delays	Requires extra time or rushes through questions	Unable to complete within given time
Question Interpretation	10%	Interprets all questions correctly and responds appropriately	Misinterprets a few questions	Often misinterprets questions	Unable to understand questions